

ENG301 - Computer Vision

Assignment 2

Demonstration deadline: 10 April 2025

Report submission deadline: 11:59 pm, 13 April 2025

Note: For the programming task, apart from the source code, students also need to submit a **video** showing the tracking result.

A. Individual tasks

A1. Theory

Consider an object moving on the image plane at a constant acceleration $(a_x, a_y) = (30, 20)$ pixel/s². This acceleration is subject to Gaussian noise with zero mean and variance of $(\sigma_{ax}^2, \sigma_{ay}^2) = (5, 4)$.

A detection algorithm was developed capable of measuring its x and y positions with zero mean and variance of (2,1).

A Kalman filter is used to estimate the state of the object including its position and velocity, given that the sampling time $T = 0.5$ s, the initial position and velocity of the object is $(x_0, y_0) = (0, 0)$ and (v_{x0}, v_{y0})

$= (0, 0)$ with the error covariance matrix $P_0 = \begin{bmatrix} 5 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 10 & 0 \\ 0 & 0 & 0 & 10 \end{bmatrix}$.

- a) Write the motion and measurement equations for this system and identify the matrices necessary for the Kalman filter, i.e., A, B, Q, H, R.

Note:

- Noise is independent.
 - The standard deviation of position is defined as the standard deviation of acceleration σ_a multiplied by $T^2/2$
 - The standard deviation of the velocity is defined as the standard deviation of the acceleration σ_a multiplied by T .
- b) Apply the Kalman filter to estimate the position and velocity of the object for the two first iterations given the measurements returned by the algorithm are as follows:

k	1	2
x	5	2
y	20	12

Note: Students can use a Python script to help with their calculation

A2. Programming

In your Assignment 1, you have developed a program to detect the center of the eye pupil. In this assignment, you will apply the Kalman filter to form a tracking system by completing the following tasks:

- a) Determine the error variance of your detection program and then the error covariance matrix R of the measurements.

- b) Implement a Kalman filter to track the center of the eye pupil. It is up to you to choose the model and initial parameters, for example, as in the Theory section or in the tutorial. However, you need to mention your choice of the model as well as your determination of the matrices for the Kalman filter.
- c) Estimate the tracking accuracy. Comment on your result.

B. Group task

Work in groups of two students to carry out the following tasks:

- a) Using the cameras provided in class to make a simple stereo system and record videos from that system.
- b) Write a program to compute the depth information from the stereo system. Export the depth map together with the color images to a point cloud file (.ply) that can be visualized using OpenCV or 3D software such as MeshLab.
- c) Write a program to use that depth information to detect the distance from the camera to the object. Generate a warning if the object is close to the camera, for example, < 50 cm.